Unit 8 Gases Worksheet #1 – Kinetic Molecular Theory and Gas Relationships Name: _____

1. One of the postulates of KMT states that collisions are perfectly elastic. In your own words, describe what this means about the collision.

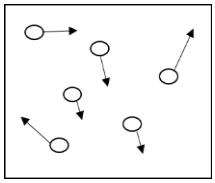
2. Gases are easily compressible whereas solids and liquids are not. Explain why.

3. How does a gas create pressure on the surroundings?

4. In the space provided below, determine if each of the variables have an inverse or direct relationship

   a. Pressure vs Volume: _____
   b. Pressure vs Temperature: _____
   c. Temperature vs Volume: _____
   d. # Moles vs Pressure: _____

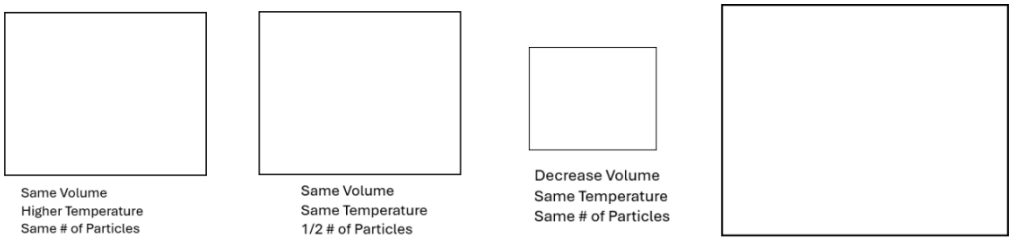
5. A sample of gas is held in a sealed container provided by the diagram below.



   a. What do the arrows indicate in the diagram provided?

   b. Why are some arrows longer than others?

   c. Four boxes are provided below. Each box is a set of conditions that have been changed about the original diagram above. In each box, draw how the conditions will affect the particles in the gas sample.



6. Perform the pressure or temperature conversion for each problem below.
   1 atm = 760 torr = 760 mmHg = 101.3 kPa = 14.7 psi K = °C + 273

   a. Convert 35.0°C to kelvin
   b. Convert 1.25 atm to kPa
   c. Convert 38.0 psi to atm

Unit 8 Gases Worksheet #2 – Combined Gas Law Name: _____

1) If I initially have a gas at a pressure of 11 atm, a volume of 23 liters, and a temperature of 200 K, and then I raise the pressure to 13 atm and increase the temperature to 300 K, what is the new volume of the gas?

2) A gas takes up a volume of 18 liters, has a pressure of 2.3 atm, and a temperature of 299 K. If I raise the temperature to 355 K and lower the pressure to 1.5 atm, what is the new volume of the gas?

3) A gas that has a volume of 31 liters, a temperature of 45^0C, and an unknown pressure has its volume increased to 36 liters and its temperature decreased to 35^0C. If I measure the pressure after the change to be 2.0 atm, what was the original pressure of the gas?

4) A gas has a temperature of 13 ^{0}C, and a volume of 4.1 liters. Calculate the new temperature if the volume of the gas is reduced to 2.5 liters assuming the pressure stays constant.

5) If I have 18 liters of gas at a temperature of 67 ^{0}C and a pressure of 78.89 atm, what will be the pressure of the gas if I raise the temperature to 94 ^{0}C and decrease the volume to 12 liters?

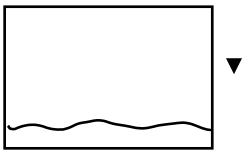
Unit 8 Gases Worksheet #3 – Ideal Gas Law

1. A typical human lung has a capacity of 3.90 liters. Assuming the lung has a pressure of 1.00 atm and a temperature of 40.0°C, calculate the number of moles of gas present in a fully inflated lung.

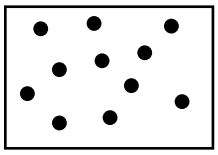
2. A sample of nitrogen gas contains 0.921 moles at a pressure of 1.38 atm and a temperature of 316 K. What volume does this gas sample occupy?

3. A sample of chlorine gas occupies a volume of 35.5 L at a pressure of 790 mmHg and a temperature of 100.0°C. Calculate the number of grams of chlorine present in that sample.

4. A student obtains an unknown liquid, Substance A, that readily evaporates into a gas. The student's goal is to set up an experiment to determine the molar mass of this unknown gas. The diagram below shows the experimental setup.



5 grams of Substance A



_____grams of Substance A

The particles are just to show you that substance A has turned into a gas. You should not use the number of particles present in any of your calculations or answers.

   a. The student allows the liquid to evaporate into the container. If all liquid has evaporated, how many grams of gas are present in the container? Assume that the only gas present in the container is Substance A. Write your answer below the box that contains the gas.

   b. The student wishes to calculate the molar mass of substance A. What are the units for molar mass? _____

   c. The container above has a volume of 5.00 liters and a temperature of 25.0°C. The pressure produced by the gas is measured to be 1.25 atm. Calculate the number of moles of gas that are present in the container.

   d. Using your information given and what you calculated in part C, determine the molar mass of substance A.

   e. The student repeats the experiment but uses 10.0 grams of substance A. Do you expect the calculated molar mass to be higher, lower, or the same as you calculated in part D

Unit 8 Gases Worksheet #4 – Ideal Gas Law Stoichiometry Name: _____

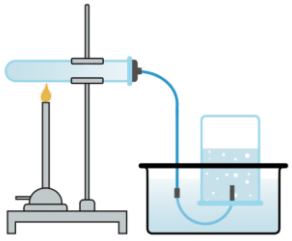
1. Gasoline is mostly made of octane(C_8H_{18}) which can be combusted in the presence of oxygen gas to produce carbon dioxide and water.
   a. Write the balanced chemical equation below for the combustion of octane.

   b. A gallon of gasoline has a mass of roughly 4000.0 grams. How many liters of CO_2 will be produced at a temperature of 21.0°C and pressure of 1.00 atm if 1 gallon of gasoline is combusted.

   c. Assuming the same temperature in pressure as part B, how many liters of oxygen gas will be consumed from the combustion of 1 gallon of gasoline?

   d. The enthalpy of combustion(ΔH) of octane is -5,460 kJ/mol rxn. Calculate the amount of energy released in the combustion of 1 gallon of octane.

2. A student is interested in performing the following experiment to collect the gas produced. The diagram to the right shows the experimental setup.



   Reaction: ____$NaN_{3\,(s)}$ → ____$Na_{(s)}$ + ____$N_{2(g)}$

   a. On the diagram provided, label where the reaction is occurring and where the gas is being collected.
   b. Sodium azide exists as a solid at room temperature. Using your knowledge of types of compounds, explain why it is found as a solid at this temperature.

   c. 10.0 grams of solid sodium azide (NaN_3) are decomposed and the resulting gas is collected. The pressure of the gas is measured to be 0.98 atm and temperature 32.0°C. Calculate the volume, in mL, of the N_2 gas produced.

   d. The reaction above is utilized in many airbags due to the speed at which nitrogen gas is produced. An airbag typically needs roughly 75.0 liters of nitrogen gas to fully inflate. How many grams of sodium azide would need to be decomposed to inflate an airbag if the gas is as a pressure of 1.0 atm and a temperature of 350 Kelvin.